(*Hint:* Show how to obtain the Bernoulli trials in $A'$ by an experiment involving the trials of $A$, and use the result of Exercise C.3-7.)

## ★ C.5 The tails of the binomial distribution

The probability of having at least, or at most, $k$ successes in $n$ Bernoulli trials, each with probability $p$ of success, is often of more interest than the probability of having exactly $k$ successes. In this section, we investigate the ***tails*** of the binomial distribution: the two regions of the distribution $b(k; n, p)$ that are far from the mean $np$. We'll prove several important bounds on (the sum of all terms in) a tail.

We first provide a bound on the right tail of the distribution $b(k; n, p)$. To determine bounds on the left tail, simply invert the roles of successes and failures.

***Theorem C.2***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$. Let $X$ be the random variable denoting the total number of successes. Then for $0 \le k \le n$, the probability of at least $k$ successes is

$$
\begin{aligned}
\Pr\{X \ge k\} &= \sum_{i=k}^{n} b(i; n, p) \\
&\le \binom{n}{k} p^k .
\end{aligned}
$$

***Proof*** For $S \subseteq \{1, 2, \ldots, n\}$, let $A_S$ denote the event that the $i$th trial is a success for every $i \in S$. Since $\Pr\{A_S\} = p^k$, where $|S| = k$, we have

$$
\begin{aligned}
\Pr\{X \ge k\} &= \Pr\{\text{there exists } S \subseteq \{1,2,\ldots,n\} : |S| = k \text{ and } A_S\} \\
&= \Pr\left\{ \bigcup_{S \subseteq \{1,2,\ldots,n\}:|S|=k} A_S \right\} \\
&\le \sum_{S \subseteq \{1,2,\ldots,n\}:|S|=k} \Pr\{A_S\} \qquad \text{(by inequality (C.21) on page 1190)} \\
&= \binom{n}{k} p^k .
\end{aligned}
$$

■

The following corollary restates the theorem for the left tail of the binomial distribution. In general, we'll leave it to you to adapt the proofs from one tail to the other.

***Corollary C.3***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$. If $X$ is the random variable denoting the total number of successes, then for $0 \le k \le n$, the probability of at most $k$ successes is

$$\begin{aligned}\Pr\{X \le k\} &= \sum_{i=0}^{k} b(i;n,p) \\ &\le \binom{n}{n-k}(1-p)^{n-k} \\ &= \binom{n}{k}(1-p)^{n-k} .\end{aligned}$$

■

Our next bound concerns the left tail of the binomial distribution. Its corollary shows that, far from the mean, the left tail diminishes exponentially.

***Theorem C.4***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$ and failure with probability $q = 1 - p$. Let $X$ be the random variable denoting the total number of successes. Then for $0 < k < np$, the probability of fewer than $k$ successes is

$$\begin{aligned}\Pr\{X < k\} &= \sum_{i=0}^{k-1} b(i;n,p) \\ &< \frac{kq}{np-k} b(k;n,p) .\end{aligned}$$

***Proof*** We bound the series $\sum_{i=0}^{k-1} b(i;n,p)$ by a geometric series using the technique from Section A.2, page 1147. For $i = 1, 2, \ldots, k$, equation (C.45) gives

$$
\begin{aligned}
\frac{b(i-1;n,p)}{b(i;n,p)} &= \frac{iq}{(n-i+1)p} \\
&< \frac{iq}{(n-i)p} \\
&\le \frac{kq}{(n-k)p} .
\end{aligned}
$$

If we let

$$
\begin{aligned}
x &= \frac{kq}{(n-k)p} \\
&< \frac{kq}{(n-np)p} \\
&= \frac{kq}{nqp} \\
&= \frac{k}{np} \\
&< 1 ,
\end{aligned}
$$

it follows that

b$(i-1; n, p) < xb(i; n, p)$

for $0 < i \le k$. Iteratively applying this inequality $k - i$ times gives

$b(i; n, p) < x^{k-i} b(k; n, p)$

for $0 \le i < k$, and hence

$$
\begin{aligned}
\sum_{i=0}^{k-1} b(i;n,p) &< \sum_{i=0}^{k-1} x^{k-i} b(k;n,p) \\
&< b(k;n,p) \sum_{i=1}^{\infty} x^i \\
&= \frac{x}{1-x} b(k;n,p) \\
&= \frac{kq/((n-k)p)}{((n-k)p-kq)/((n-k)p)} b(k;n,p) \\
&= \frac{kq}{np-kp-kq} b(k;n,p) \\
&= \frac{kq}{np-k} b(k;n,p) .
\end{aligned}
$$

■

***Corollary C.5***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$ and failure with probability $q = 1 - p$. Then for $0 < k \le np/2$, the probability of fewer than $k$ successes is less than half the probability of fewer than $k + 1$ successes.

***Proof*** Because $k \le np/2$, we have

$$\begin{aligned} \frac{kq}{np-k} &\le \frac{(np/2)q}{np-(np/2)} \\ &= \frac{(np/2)q}{np/2} \\ &\le 1 , \end{aligned} \tag{C.46}$$

since $q \le 1$. Letting $X$ be the random variable denoting the number of successes, Theorem C.4 and inequality (C.46) imply that the probability of fewer than $k$ successes is

$$\Pr\{X < k\} = \sum_{i=0}^{k-1} b(i;n,p) < b(k;n,p) .$$

Thus we have

$$\begin{aligned} \frac{\Pr\{X < k\}}{\Pr\{X < k+1\}} &= \frac{\sum_{i=0}^{k-1} b(i;n,p)}{\sum_{i=0}^{k} b(i;n,p)} \\ &= \frac{\sum_{i=0}^{k-1} b(i;n,p)}{\sum_{i=0}^{k-1} b(i;n,p) + b(k;n,p)} \\ &< 1/2 , \end{aligned}$$

since $\sum_{i=0}^{k-1} b(i;n,p) < b(k;n,p)$.

■

Bounds on the right tail follow similarly. Exercise C.5-2 asks you to prove them.

***Corollary C.6***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$. Let $X$ be the random variable denoting the total number of successes. Then for $np < k < n$, the probability of more than $k$ successes is

$$
\begin{aligned}
\Pr\{X > k\} &= \sum_{i=k+1}^{n} b(i; n, p) \\
&< \frac{(n-k)p}{k-np} b(k; n, p) .
\end{aligned}
$$

■

***Corollary C.7***

Consider a sequence of $n$ Bernoulli trials, where success occurs with probability $p$ and failure with probability $q = 1 - p$. Then for $(np + n)/2 < k < n$, the probability of more than $k$ successes is less than half the probability of more than $k - 1$ successes.

■

The next theorem considers $n$ Bernoulli trials, each with a probability $p_i$ of success, for $i = 1, 2, \ldots, n$. As the subsequent corollary shows, we can use the theorem to provide a bound on the right tail of the binomial distribution by setting $p_i = p$ for each trial.

***Theorem C.8***

Consider a sequence of $n$ Bernoulli trials, where in the $i$th trial, for $i = 1, 2, \ldots, n$, success occurs with probability $p_i$ and failure occurs with probability $q_i = 1 - p_i$. Let $X$ be the random variable describing the total number of successes, and let $\mu = \mathrm{E}[X]$. Then for $r > \mu$,

$$\Pr\{X - \mu \geq r\} \leq \left(\frac{\mu e}{r}\right)^r .$$

***Proof*** Since for any $\alpha > 0$, the function $e^{\alpha x}$ strictly increases in $x$,

$$\Pr\{X - \mu \geq r\} = \Pr\left\{e^{\alpha(X-\mu)} \geq e^{\alpha r}\right\} , \tag{C.47}$$

where we will determine $\alpha$ later. Using Markov's inequality (C.34), we obtain

$$\Pr\left\{e^{\alpha(X-\mu)} \geq e^{\alpha r}\right\} \leq \mathrm{E}\left[e^{\alpha(X-\mu)}\right] e^{-\alpha r} . \tag{C.48}$$

The bulk of the proof consists of bounding $\mathrm{E}[e^{\alpha(X-\mu)}]$ and substituting a suitable value for $\alpha$ in inequality (C.48). First, we evaluate $\mathrm{E}[e^{\alpha(X-\mu)}]$. Using the technique of indicator random variables (see

Section 5.2), let $X_i$ = I {the $i$th Bernoulli trial is a success} for $i$ = 1, 2, … , $n$. That is, $X_i$ is the random variable that is 1 if the $i$th Bernoulli trial is a success and 0 if it is a failure. Thus, we have

$$X = \sum_{i=1}^{n} X_i ,$$

and by linearity of expectation,

$$\mu = \mathrm{E}[X] = \mathrm{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathrm{E}[X_i] = \sum_{i=1}^{n} p_i ,$$

which implies

$$X - \mu = \sum_{i=1}^{n} (X_i - p_i) .$$

To evaluate $\mathrm{E}[e^{\alpha(X-\mu)}]$, we substitute for $X - \mu$, obtaining

$$\begin{aligned}
\mathrm{E}\left[e^{\alpha(X-\mu)}\right] &= \mathrm{E}\left[e^{\alpha\sum_{i=1}^{n}(X_i-p_i)}\right] \\
&= \mathrm{E}\left[\prod_{i=1}^{n} e^{\alpha(X_i-p_i)}\right] \\
&= \prod_{i=1}^{n} \mathrm{E}\left[e^{\alpha(X_i-p_i)}\right] ,
\end{aligned}$$

which follows from equation (C.27), since the mutual independence of the random variables $X_i$ implies the mutual independence of the random variables $e^{\alpha(X_i-p_i)}$ (see Exercise C.3-5). By the definition of expectation,

$$\begin{aligned}
\mathrm{E}\left[e^{\alpha(X_i-p_i)}\right] &= e^{\alpha(1-p_i)} p_i + e^{\alpha(0-p_i)} q_i \\
&= p_i e^{\alpha q_i} + q_i e^{-\alpha p_i} \\
&\le p_i e^{\alpha} + 1 \qquad\qquad (C.49) \\
&\le \exp(p_i e^{\alpha}) ,
\end{aligned}$$

where $\exp(x)$ denotes the exponential function: $\exp(x) = e^x$. (Inequality (C.49) follows from the inequalities $\alpha > 0$, $q_i \le 1$, $e^{\alpha q_i} \le e^{\alpha}$, and $e^{-\alpha p_i} \le 1$. The last line follows from inequality (3.14) on page 66.) Consequently,

$$
\begin{aligned}
\mathrm{E}\left[e^{\alpha(X-\mu)}\right] &= \prod_{i=1}^{n} \mathrm{E}\left[e^{\alpha(X_i-p_i)}\right] \\
&\le \prod_{i=1}^{n} \exp(p_i e^{\alpha}) \\
&= \exp\left(\sum_{i=1}^{n} p_i e^{\alpha}\right) \\
&= \exp(\mu e^{\alpha}) \, , \qquad \text{(C.50)}
\end{aligned}
$$

since $\mu = \sum_{i=1}^{n} p_i$. Therefore, from equation (C.47) and inequalities (C.48) and (C.50), it follows that

$$
\Pr\{X - \mu \ge r\} \le \exp(\mu e^{\alpha} - \alpha r) \, . \qquad \text{(C.51)}
$$

Choosing $\alpha = \ln(r/\mu)$ (see Exercise C.5-7), we obtain

$$
\begin{aligned}
\Pr\{X - \mu \ge r\} &\le \exp(\mu e^{\ln(r/\mu)} - r\ln(r/\mu)) \\
&= \exp(r - r\ln(r/\mu)) \\
&= \frac{e^r}{(r/\mu)^r} \\
&= \left(\frac{\mu e}{r}\right)^r .
\end{aligned}
$$

■

When applied to Bernoulli trials in which each trial has the same probability of success, Theorem C.8 yields the following corollary bounding the right tail of a binomial distribution.

***Corollary C.9***

Consider a sequence of $n$ Bernoulli trials, where in each trial success occurs with probability $p$ and failure occurs with probability $q = 1 - p$. Then for $r > np$,

$$
\begin{aligned}
\Pr\{X - np \ge r\} &= \sum_{k=\lceil np+r \rceil}^{n} b(k; n, p) \\
&\le \left(\frac{npe}{r}\right)^r .
\end{aligned}
$$

***Proof*** By equation (C.41), we have $\mu = \mathrm{E}[X] = np$.

■

**Exercises**

★ *C.5-1*

Which is more likely: getting exactly $n$ heads in $2n$ flips of a fair coin, or $n$ heads in $n$ flips of a fair coin?

★ *C.5-2*

Prove Corollaries C.6 and C.7.

★ *C.5-3*

Show that

$$\sum_{i=0}^{k-1} \binom{n}{i} a^i < (a+1)^n \frac{k}{na - k(a+1)} b(k; n, a/(a+1))$$

for all $a > 0$ and all $k$ such that $0 < k < na/(a + 1)$.

★ *C.5-4*

Prove that if $0 < k < np$, where $0 < p < 1$ and $q = 1 - p$, then

$$\sum_{i=0}^{k-1} p^i q^{n-i} < \frac{kq}{np-k} \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k} .$$

★ *C.5-5*

Use Theorem C.8 to show that

$$\Pr\{\mu - X \geq r\} \leq \left(\frac{(n-\mu)e}{r}\right)^r$$

for $r > n - \mu$. Similarly, use Corollary C.9 to show that

$$\Pr\{np - X \geq r\} \leq \left(\frac{nqe}{r}\right)^r$$

for $r > n - np$.

★ *C.5-6*

Consider a sequence of $n$ Bernoulli trials, where in the $i$th trial, for $i = 1, 2, \ldots, n$, success occurs with probability $p_i$ and failure occurs with probability $q_i = 1 - p_i$. Let $X$ be the random variable describing the total number of successes, and let $\mu = E[X]$. Show that for $r \geq 0$,

$$\Pr\{X - \mu \geq r\} \leq e^{-r^2/2n} .$$

(*Hint:* Prove that $p_i e^{\alpha q_i} + q_i e^{-\alpha p_i} \leq e^{\alpha^2/2}$. Then follow the outline of the proof of Theorem C.8, using this inequality in place of inequality (C.49).)

★ ***C.5-7***

Show that choosing $\alpha = \ln(r/\mu)$ minimizes the right-hand side of inequality (C.51).

## Problems

### *C-1 The Monty Hall problem*

Imagine that you are a contestant in the 1960s game show *Let's Make a Deal*, hosted by emcee Monty Hall. A valuable prize is hidden behind one of three doors and comparatively worthless prizes behind the other two doors. You will win the valuable prize, typically an automobile or other expensive product, if you select the correct door. After you have picked one door, but before the door has been opened, Monty, who knows which door hides the automobile, directs his assistant Carol Merrill to open one of the other doors, revealing a goat (not a valuable prize). He asks whether you would like to stick with your current choice or to switch to the other closed door. What should you do to maximize your chances of winning the automobile and not the other goat?

The answer to this question—stick or switch?—has been heavily debated, in part because the problem setup is ambiguous. We'll explore different subtle assumptions.

***a.*** Suppose that your first pick is random, with probability 1/3 of choosing the right door. Moreover, you know that Monty always gives every contestant (and will give you) the opportunity to switch. Prove that it is better to switch than stick. What is your probability of winning the automobile?

This answer is the one typically given, even though the original statement of the problem rarely mentions the assumption that Monty *always* offers the contestant the opportunity to switch. But, as the remainder of this problem will elucidate, your best strategy may be different if this unstated assumption does not hold. In fact, in the real game show, after a contestant picked a door, Monty sometimes simply asked Carol to open the door that the contestant had chosen.

Let's model the interactions between you and Monty as a probabilistic experiment, where you both employ randomized strategies. Specifically, after you pick a door, Monty offers you the opportunity to switch with probability $p_{\text{right}}$ if you picked the right door and with probability $p_{\text{wrong}}$ if you picked the wrong door. Given the opportunity to switch, you randomly choose to switch with probability $p_{\text{switch}}$. For example, if Monty always offers you the opportunity to switch, then his strategy is given by $p_{\text{right}} = p_{\text{wrong}} = 1$. If you always switch, then your strategy is given by $p_{\text{switch}} = 1$.

The game can now be viewed as an experiment consisting of five steps:

1. You pick a door at random, choosing the automobile (right) with probability 1/3 or a goat (wrong) with probability 2/3.
2. Carol opens one of the two closed doors, revealing a goat.
3. Monty offers you the opportunity to switch with probability $p_{\text{right}}$ if your choice is right and with probability $p_{\text{wrong}}$ if your choice is wrong.
4. If Monty makes you an offer in step 3, you switch with probability $p_{\text{switch}}$.
5. Carol opens the door you've chosen, revealing either an automobile (you win) or a goat (you lose).

Let's now analyze this game and understand how the choices of $p_{\text{right}}$, $p_{\text{wrong}}$, and $p_{\text{switch}}$ influence the probability of winning.

***b.*** What are the six outcomes in the sample space for this game? Which outcomes correspond to you winning the automobile? What are the probabilities in terms of $p_{\text{right}}$, $p_{\text{wrong}}$, and $p_{\text{switch}}$ of each outcome? Organize your answers into a table.

***c.*** Use the results of your table (or other means) to prove that the probability of winning the automobile is

$$\frac{1}{3}(2\,p_{\text{wrong}}\,p_{\text{switch}} - p_{\text{right}}\,p_{\text{switch}} + 1)\,.$$

Suppose that Monty knows the probability $p_{\text{switch}}$ that you switch, and his goal is to minimize your chance of winning.

***d.*** If $p_{\text{switch}} > 0$ (you switch with a positive probability), what is Monty's best strategy, that is, his best choice for $p_{\text{right}}$ and $p_{\text{wrong}}$?

***e.*** If $p_{\text{switch}} = 0$ (you always stick), argue that all of Monty's possible strategies are optimal for him.

Suppose that now Monty's strategy is fixed, with particular values for $p_{\text{right}}$ and $p_{\text{wrong}}$.

***f.*** If you know $p_{\text{right}}$ and $p_{\text{wrong}}$, what is your best strategy for choosing your probability $p_{\text{switch}}$ of switching as a function of $p_{\text{right}}$ and $p_{\text{wrong}}$?

***g.*** If you don't know $p_{\text{right}}$ and $p_{\text{wrong}}$, what choice of $p_{\text{switch}}$ maximizes the minimum probability of winning over all the choices of $p_{\text{right}}$ and $p_{\text{wrong}}$?

Let's return to the original problem as stated, where Monty has given you the option of switching, but you have no knowledge of Monty's possible motivations or strategies.

***h.*** Argue that the conditional probability of winning the automobile given that Monty offers you the opportunity to switch is

$$\frac{p_{\text{right}} - p_{\text{right}}\,p_{\text{switch}} + 2\,p_{\text{wrong}}\,p_{\text{switch}}}{p_{\text{right}} + 2\,p_{\text{wrong}}}\,. \qquad \text{(C.52)}$$

Explain why $p_{\text{right}} + 2p_{\text{wrong}} \neq 0$.

***i.*** What is the value of expression (C.52) when $p_{\text{switch}} = 1/2$? Show that choosing $p_{\text{switch}} < 1/2$ or $p_{\text{switch}} > 1/2$ allows Monty to select values for $p_{\text{right}}$ and $p_{\text{wrong}}$ that yield a lower value for expression (C.52) than choosing $p_{\text{switch}} = 1/2$.

***j.*** Suppose that you don't know Monty's strategy. Explain why choosing to switch with probability 1/2 is a good strategy for the original problem as stated. Summarize what you have learned overall from this problem.

### ***C-2 Balls and bins***

This problem investigates the effect of various assumptions on the number of ways of placing $n$ balls into $b$ distinct bins.

***a.*** Suppose that the $n$ balls are distinct and that their order within a bin does not matter. Argue that the number of ways of placing the balls in the bins is $b^n$.

***b.*** Suppose that the balls are distinct and that the balls in each bin are ordered. Prove that there are exactly $(b + n - 1)!/(b - 1)!$ ways to place the balls in the bins. (*Hint:* Consider the number of ways of arranging $n$ distinct balls and $b - 1$ indistinguishable sticks in a row.)

***c.*** Suppose that the balls are identical, and hence their order within a bin does not matter. Show that the number of ways of placing the balls in the bins is $\binom{b+n-1}{n}$. (*Hint:* Of the arrangements in part (b), how many are repeated if the balls are made identical?)

***d.*** Suppose that the balls are identical and that no bin may contain more than one ball, so that $n \le b$. Show that the number of ways of placing the balls is $\binom{b}{n}$.

***e.*** Suppose that the balls are identical and that no bin may be left empty. Assuming that $n \ge b$, show that the number of ways of placing the balls is $\binom{n-1}{b-1}$.

## Appendix notes

The first general methods for solving probability problems were discussed in a famous correspondence between B. Pascal and P. de Fermat, which began in 1654, and in a book by C. Huygens in 1657. Rigorous probability theory began with the work of J. Bernoulli in 1713 and A. De Moivre in 1730. Further developments of the theory were provided by P.-S. Laplace, S.-D. Poisson, and C. F. Gauss.

Sums of random variables were originally studied by P. L. Chebyshev and A. A. Markov. A. N. Kolmogorov axiomatized probability theory in 1933. Chernoff [91] and Hoeffding [222] provided bounds on the tails of distributions. Seminal work in random combinatorial structures was done by P. Erdős.

Knuth [259] and Liu [302] are good references for elementary combinatorics and counting. Standard textbooks such as Billingsley [56], Chung [93], Drake [125], Feller [139], and Rozanov [390] offer comprehensive introductions to probability.

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[1] For a general probability distribution, there may be some subsets of the sample space $S$ that are not considered to be events. This situation usually arises when the sample space is uncountably infinite. The main requirement for what subsets are events is that the set of events of a sample space must be closed under the operations of taking the complement of an event, forming the union of a finite or countable number of events, and taking the intersection of a finite or countable number of events. Most of the probability distributions we see in this book are over finite or countable sample spaces, and we generally consider all subsets of a sample space to be events. A notable exception is the continuous uniform probability distribution, which we'll see shortly.